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Analog Guitar Effects Processing Console

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Abstract

This project presents the **Analog Guitar Effects Processing Console**, a compact device integrating four analog sub-circuits for guitar tone modification: **Pre-Amplifier**, **Equalizer**, **Distortion**, and **Tremolo**. The system utilizes modular architecture for reliability. Validation was conducted via LTSpice simulation and hardware testing. Hardware validation included FFT analysis to confirm the active equalizer's effective boost/cut ranges and oscilloscope and digital multimeter readings for the rest of the circuit. The console successfully demonstrates a functional, portable, and versatile analog audio processing unit.

Abbreviations and Acronyms

FFT Fast Fourier Transform

SR Slew Rate

DC Direct Current

AC Alternating Current

PCB Printed Circuit Board

LFO Low Frequency Oscillator

AWGN Additive White Gaussian Noise

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1 Introduction and Functionality

This project aims to introduce a compact, custom audio console intended to for use with analog input devices, primarily the electric guitar. The output is designed to be fed directly into a separate power amplifier, allowing musicians to pre-set their entire core tone and effects chain in a single, portable device.

The console is engineered for live performance reliability and portability, requiring only initial setup and minor, on-the-demand adjustments.

This console consists of four main parts:

- Pre-Amplifier: For initial gain
- Equalizer: To amplify different frequency ranges (High, Mid, Treble)
- Tremolo: For Tremolo effect. The frequency and Depth can be controlled.
- Distortion: Distortion effect can be controlled



Figure 1: Analog Guitar Effects Unit

2 System Architecture

In this section, we'll go through the circuitry system in an abstract manner. We'll also mention the equations that help in identifying key parameters related to the product.

2.1 System-Level Abstraction of the Device

In Figure 2, we can see how the four systems are interconnected with input and output signals and how they are powered up and optionally bypassed.

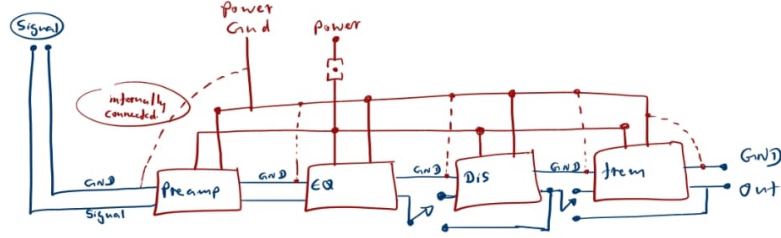


Figure 2: Flowgraph of the Circuitry

2.2 Equations

- Pre-Amplifier Voltage Gain (A_v):

$$A_v = 1 + \frac{R_f}{R_i}$$

- Baxandall EQ Center Frequency for the mid-range control:

$$\omega_c = \frac{1}{RC}$$

- Tremolo Timing (NE555 Astable Mode):

$$f = \frac{1.44}{(R_A + 2R_B)C}$$

$$\text{Duty Cycle(\%)} = \frac{R_A + R_B}{R_A + 2R_B} \times 100$$

2.3 Sub-Circuits

The device contains four separate sub-circuits—one for each effect—so that each module can be built, tested, and verified independently before integration. This simplifies debugging, as any failure or undesired behavior can be quickly isolated. Also, if any fault occurs during a live performance (eg: Tremolo Board), it can be quickly swapped out without having to replace or repair the entire console circuit.

3 Component Selection

In this section, we will illustrate some of the decisions we made in selecting components and the reasons behind them.

- TL071CP / TL082CP

Requirements:

- Power dissipation at idle: 0.4 mW
- Op-amp bandwidth: 80 Hz to 3 kHz

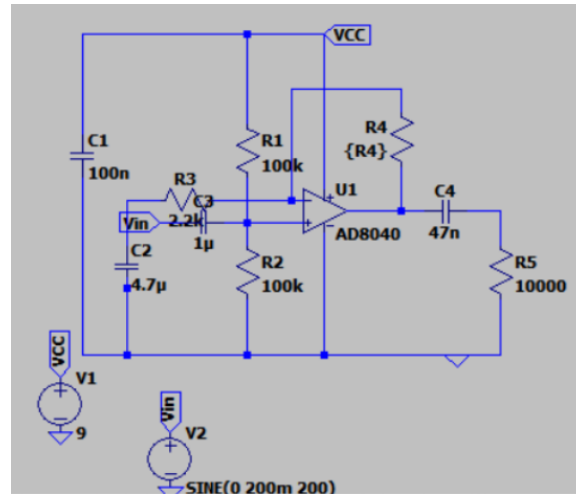


Figure 3: Pre-Amplifier

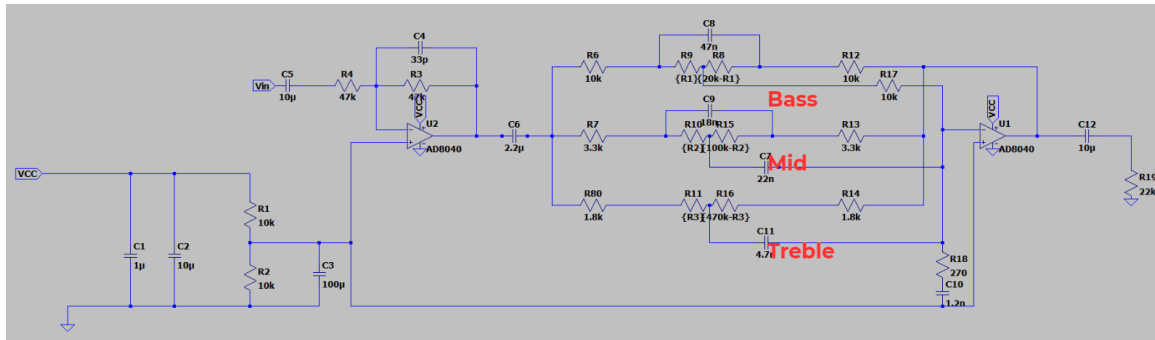


Figure 4: Equalizer

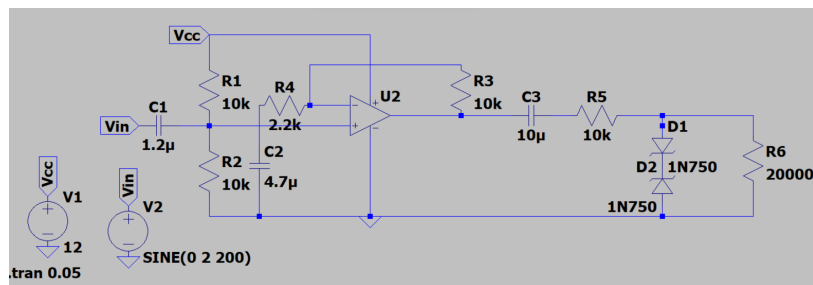


Figure 5: Distortion Circuit

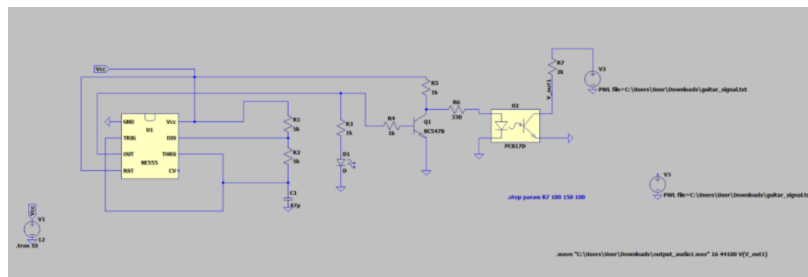


Figure 6: Tremolo Circuit

- Input amplitude:
 - * Average: 100–200 mV
 - * Maximum: 500 mV
- Expected output amplitude: 1–2 V
- Required gain-bandwidth product: 0.064 MHz
- Required slew rate (SR): 226.1 V/ms

We used popular low-noise operational amplifiers TL082CP (Dual Op-amp) and TL072CP (Single Op-amp) since they satisfy all the requirements. For the circuits that require two op-amps, such as the Equalizer, we used the dual op-amp to reduce component count.

- NE555 Timer

We used this hybrid component because of its simplicity, reliability, and stability in providing an accurate timing for tremolo effect, which is essential in live performances. For this particular application, we used Astable mode.

- Zener Diode Selection

We observed that the output of the equalizer is around $2V - 3V$. To have a good distortion effect, it should be able to produce a higher gain (for consistency in volume) to be clipped. So we used 3.3V zener diodes so that the signal can be clipped around 4V, necessitating the requirement.

- Resistors and Capacitors

Based on circuit simulations and mathematical Equations provided in Section 2.2.

4 PCB Design

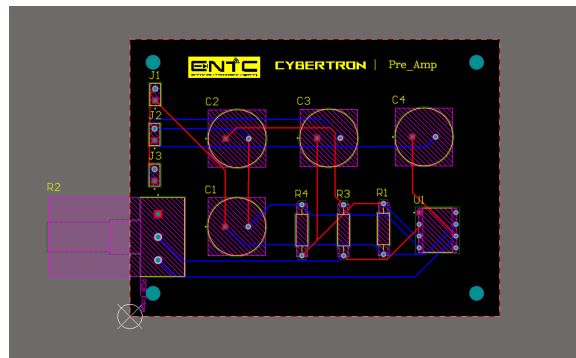


Figure 7: Pre-Amplifier

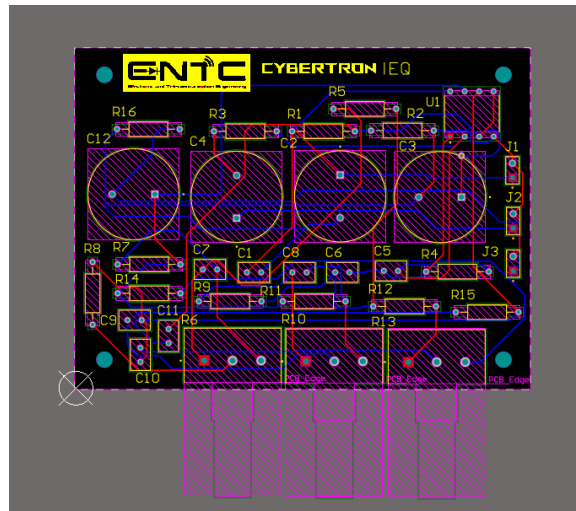


Figure 8: Equalizer

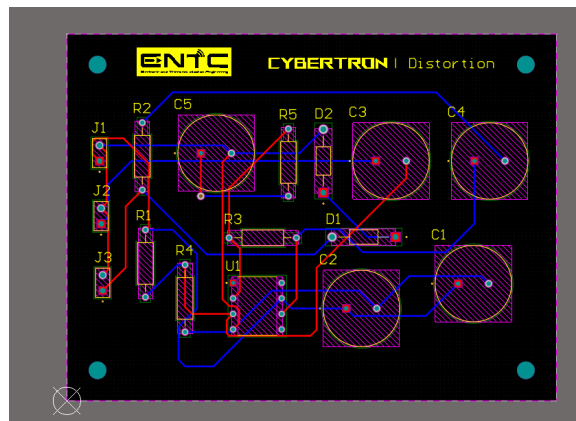


Figure 9: Distortion

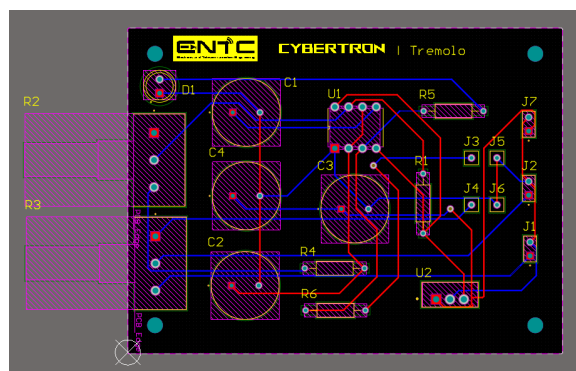


Figure 10: Tremolo

All the PCB designs are 80 mm × 60 mm in dimensions and consist of two layers. Designs are done using Altium Designer software.

5 Enclosure Design

Physical Specifications

- **Enclosure Dimensions (L × W × H):** 160 mm × 100 mm × 53 mm
- **Overall Weight:** 200 g
- **Fabrication Material:** PLA (Polylactic Acid)

SOLIDWORKS Designs

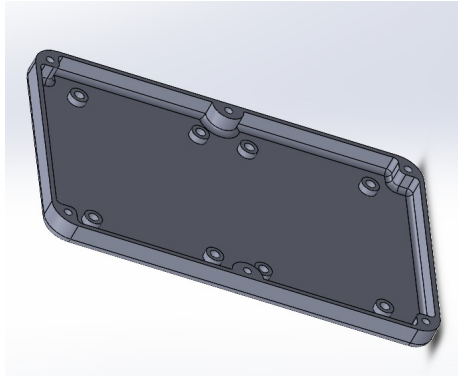


Figure 11: Bottom Lid

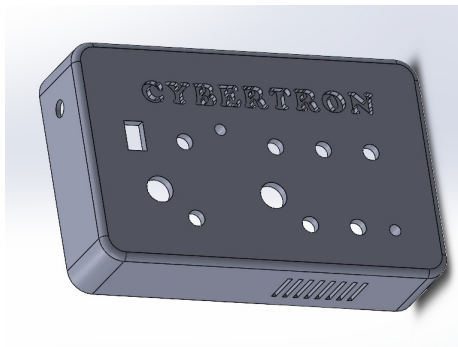


Figure 12: Top Lid

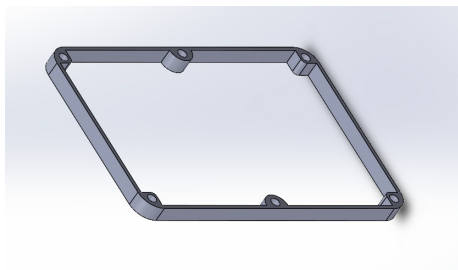


Figure 13: Inter-Layer

6 Software Simulation and Hardware Testing

This section details the verification of the console's functional blocks through circuit simulation (LTSpice) and the validation of key performance metrics using hardware testing and analysis (Fast Fourier Transform, FFT).

6.1 Simulation Results (LTSpice)

Each module's design was simulated to ensure expected electrical behavior, gain structure, frequency response, and transient stability before fabrication.

6.1.1 Pre-Amplifier Simulation

- **Goal:** Verify whether the required gain is achieved with minimal phase offset.
- **Result:** The simulation confirmed a maximum non-inverting gain of 20 dB with negligible DC offset, correctly driving the signal to the Equalizer stage.

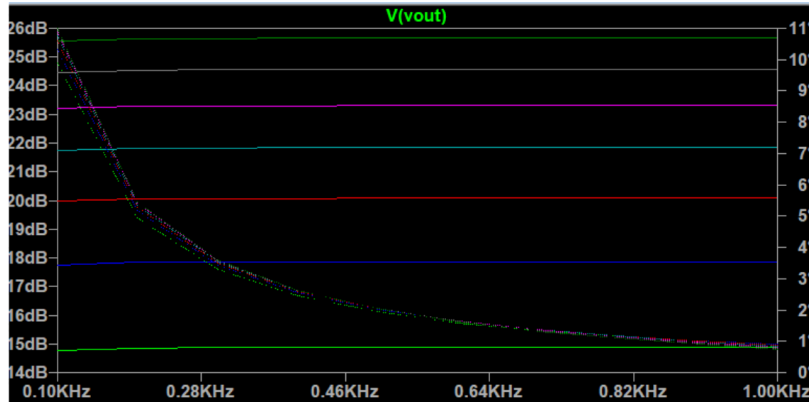


Figure 14:

- 1) — Gain Stability of pre-amp across a range of frequencies, for different values of potentiometer ($0\ \Omega$ - $20\ \Omega$)
- 2) - - Phase offset for different potentiometer values. Observation: Negligible phase offsets.

6.1.2 Equalizer Simulation

- **Goal:** Verify the center frequencies and the maximum boost/cut range of the three-band active filter.
- **Result:** Output voltage for 2V input from preamp shows that the required gain and cutoff frequencies have been achieved.

Bass: (Resistance from 0 to 20k - linearly varied)

3.45058238506	2.08457928896	1.35606133938
2.94360125065	1.83637356758	1.2171574831
2.52275240421	1.63145083189	1.10875204206
2.16973340511	1.46130150557	1.02398249507
1.86904877424	1.31373012066	0.957232445478
1.6079750061	1.18975687027	0.904615879059
1.38033860922	1.08049321175	0.864086031914
1.18064850569	0.984675467014	0.833368420601
1.00547903776	0.900928378105	0.81070381403
0.850183784962	0.830699443817	0.796066850424
0.712432652712	0.778349995613	0.802565932274
100 Hz	215 Hz	400 Hz

Figure 15: Gain variation for Bass across low frequencies

Mid: (Resistance from 0 to 100k - linearly varied)

2.06446313858	4.63471198082	4.9167907238	3.89049077034
1.93043392897	3.19697952271	3.04356324673	2.40440201759
1.73371779919	2.47358870506	2.34176981449	1.91148090363
1.5497776866	2.06271922588	1.97871845961	1.66953271627
1.39232438803	1.78894388676	1.74859631062	1.52382051945
1.25479066372	1.58137303591	1.57540041208	1.4192147851
1.131755054	1.40066945553	1.42414009571	1.32845878601
1.01475259662	1.22408783436	1.26937359571	1.23849356174
0.90420666337	1.03062272072	1.08698546886	1.12280040979
0.800554037094	0.800157159567	0.842697292566	0.944391310215
0.727868705988	0.551754504442	0.529841274023	0.640771329403
200 Hz	502 Hz	796 Hz	1261 Hz

Figure 16: Gain variation for Mid across midband frequencies

Treble: (Resistance from 0 to 470k - linearly varied)

1.2872800231	1.63248288631	2.21307766438	3.05595815182
1.21499049664	1.39783865213	1.71966248751	2.02696204185
1.10632634163	1.24805021286	1.51672083139	1.79688405991
1.04520452023	1.17974805832	1.43962007761	1.71226793528
1.00620537996	1.1387591362	1.39637964964	1.66377842426
0.977222532034	1.10926151276	1.36470085382	1.62980318069
0.951944291592	1.08342576027	1.33575612307	1.59596180916
0.925755888224	1.05532377958	1.30394744873	1.5569756031
0.89302572608	1.01648533344	1.25556218624	1.49660563469
0.846599519253	0.952425509691	1.15906077623	1.36436313391
0.832494646311	0.926437795162	1.07596975565	1.07898539305
700 Hz	1109 Hz	1758 Hz	2786 Hz

Figure 17: Gain variation for treble across high frequencies

6.1.3 Distortion Simulation

- **Goal:** Verify the clipping action of the back-to-back diode arrangement and distortion effect.
- **Result:** The transient analysis confirmed symmetrical clipping at approximately ± 4 V (Zener breakdown voltage + forward bias voltage).



Figure 18: Distortion Effect: Voltage variation with respect to time

6.1.4 Tremolo Simulation

- **Goal:** Verify the Low Frequency Oscillator (LFO) frequency range and the modulation depth control.
- **Result:** The NE555 LFO circuit achieved a frequency range from 0.5 Hz to 6.5 Hz. The VCA/Opto-coupler simulation confirmed the ability to modulate the audio signal amplitude from 0% to 90% depth.

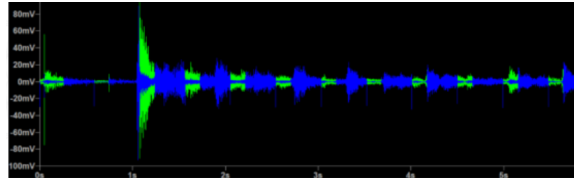


Figure 19: Tremolo Effect Simulation: Green - Input Signal, Blue - Output Signal

6.2 Hardware Testing and Validation

The physical prototype was subjected to key measurements to validate the core audio performance against specifications.

6.2.1 FFT Analysis for Equalizer Performance

- **Test:** To have a thorough understanding of frequency response, ideally, one should produce an impulse response, which isn't practical. So what the test does is provide an AWGN signal and a normal guitar strum separately and observe the FFT through an oscilloscope.
- **Goal:** Confirm the filter's response for the given frequency ranges.
- **Result:**

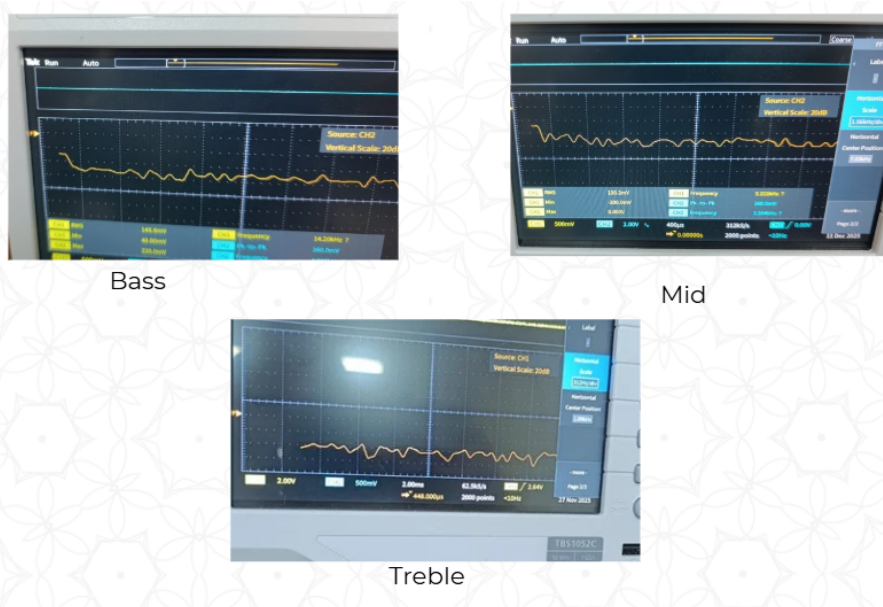


Figure 20: FFT plot validating the boost of magnitude and filtering of different frequency bands.

Explanation: Guitars naturally produce high power for low frequencies (due to higher vibrations). In addition, in the oscilloscope, it isn't easy to distinguish base frequencies and mid frequencies due to the minimum sensitivity being 1.5 kHz/div. But the treble effect demonstrates a significant observation by attenuating lower frequencies, effectively highlighting the high frequencies of the audio.

7 Conclusion & Future Works

The Analog Guitar Effects Processing Console was successfully designed and validated, realizing a compact device that integrates four essential analog effects: Pre-Amplifier, Equalizer, Distortion, and Tremolo. The system's modular, low-noise architecture was confirmed through simulation and hardware testing to meet all performance specifications, delivering a versatile and reliable multi-effect processing unit.

Future enhancements should focus on expanding the console's versatility. The modular design allows for the straightforward integration of new effects, notably the addition of a **Wah Filter** and a **Vibrato Effect** module.

8 Contribution of Group Members

Table 1: Individual Group Member Contributions

Member Name / Index	Contributions
230018T D.M.G.H. Adikaram	PCB Design, Component Selection, Assembling, Soldering
230074J H.Y.W. Bandara	Circuit Testing, PCB Design, Soldering, Assembling
230476T G.M.B. Perera	Simulation Testing, Enclosure Design, Circuit Selection, Report Documentation
230486A U.I.H. Perera	Simulation Testing, Circuit Testing, Datasheet and User Manual Documentation

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The technical staff at the Digital Laboratory of Electronic and Telecommunication Department provided necessary access to equipment and components for circuit simulation and analysis

Finally, thanks to our friends, family, and colleagues for their encouragement and support throughout this project.

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- [1] Circuit Digest. Audio Equalizer / Tone Control Circuit with Bass, Treble and Mid Frequency Control. <https://circuitdigest.com/electronic-circuits/audio-equalizer-tone-control-circuit-with-bass-treble-and-mid-frequency-control>.
- [2] Curious Motor. Simple 555 based Tremolo Effect. <https://curiousmotor.com/2020/06/04/simple-555-based-tremolo-effect/>, 2020. .

Appendix

A. Component Bill of Materials (BOM) and Estimated Cost

A list of specific components required to construct the final product, with the prices in LKR.

Component	Count	Value/Type	Unit Price	Total
IC1	1	555 Timer (NE555)	30	30
IC2	2	TL071CP	50	100
IC3	1	TL082CP	20	20
Resistor	30	10k Ω , 100k Ω , 22k Ω , etc.	5	150
Capacitor	20	1 μ F, 10 μ F, etc.	20	400
Potentiometer	7	100k Ω	40	280
LED	3	Standard Red LED	10	30
LDR	1	4mm LDR (30k Ω – 3M Ω)	20	20
PCB	4	-	200	800
AC-DC converter	1	230V AC to 12V DC Transformer	850	850
Enclosure	-	Manufactured using PLA	-	2000
Additional Expenses (Wires, Shipping, Ports, knobs etc.)	-	-	-	6000
Estimated Total Manufacturing Cost (LKR):				10680

B. Mathematical Derivation of LFO Frequency

The frequency of oscillation (f) for the 555 Timer in its astable configuration, commonly used for the Tremolo LFO, is determined by the timing resistors (R_A , R_B) and the timing capacitor (C).

The frequency is calculated using the following equation:

$$f = \frac{1.44}{(R_a + 2R_b)C}$$

$$R_a = 50 \text{ k}\Omega$$

$$R_b = 0\text{--}500 \text{ k}\Omega \text{ (variable resistor)}$$

$$C = 47 \text{ }\mu\text{F}$$

$$f_{\min} = 0.549 \text{ Hz}$$

$$f_{\max} = 6.417 \text{ Hz}$$

In a practical tremolo circuit, one of the resistors (typically R_B) is made variable (a potentiometer) to allow the user to adjust the modulation rate. For slow modulation, C_T must be large (e.g., $1\mu\text{F}$ or larger).